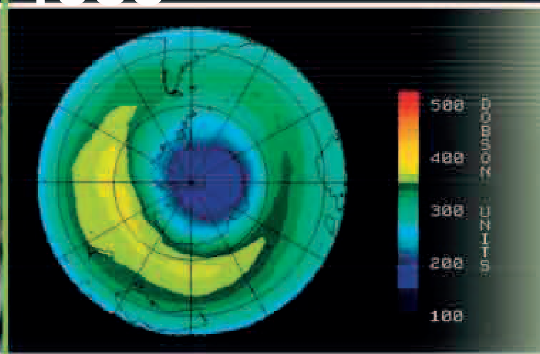
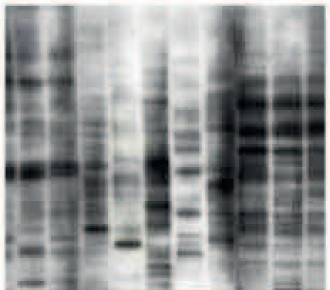


An image, constructed from satellite data, shows the ozone "hole" over Antarctica, in October 1985.

66 THE PERCENTAGE BY WHICH **OZONE** IS **DEPLETED** EACH SPRING OVER **ANTARCTICA**



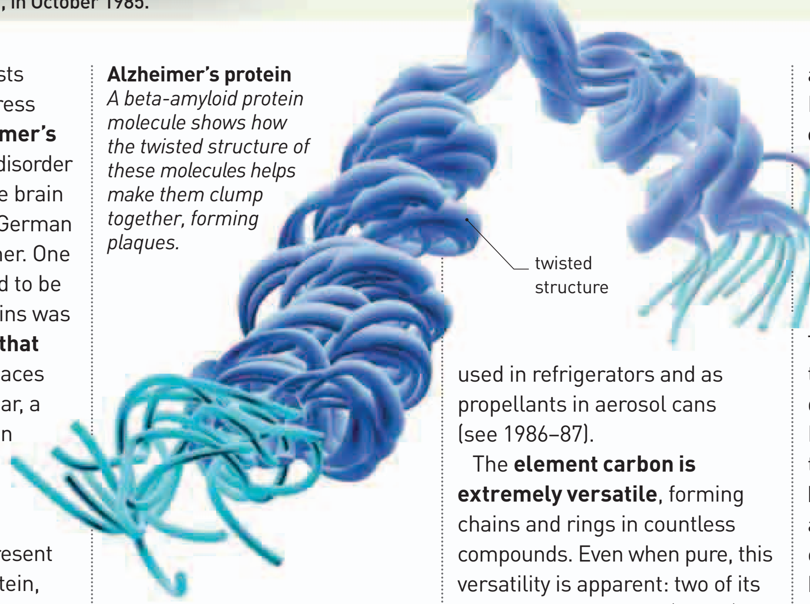
DNA PROFILING

Although DNA differs minimally between any two individuals, some sections of the genome do vary. In DNA profiling, these sections can be cut and then be arranged, in a gel, according to length. Photographs of these fragments resemble barcodes and can be used to identify an individual with a high degree of certainty.

DNA, such as blood or saliva. He later improved the sensitivity of his new procedure, by employing a technique called the **polymerase chain reaction** (PCR), which was first reported in June by American biochemist **Kary Mullis** (b.1944). This process enables molecular biochemists to **multiply small sections of DNA** almost indefinitely. PCR is a crucial element in many important DNA technologies, including DNA sequencing, cloning, and profiling.

UNTIL THE 1980s, scientists had made very little progress in **understanding Alzheimer's disease**—an age-related disorder affecting nerve cells in the brain first described in 1906 by German psychiatrist Alois Alzheimer. One phenomenon that seemed to be common in sufferers' brains was the **build-up of proteins that form "plaques"** in the spaces between neurons. This year, a team headed by Australian neuropathologist Colin Masters (b.1947) published the **first clear analysis of the protein** present in these plaques. The protein, called A-Beta amyloid, had first been described by American pathologist George Glenner (1927–95) just a year earlier. Within two years, another protein, called tubulin associated unit (tau), would also be implicated in the disease (see 1986–87).

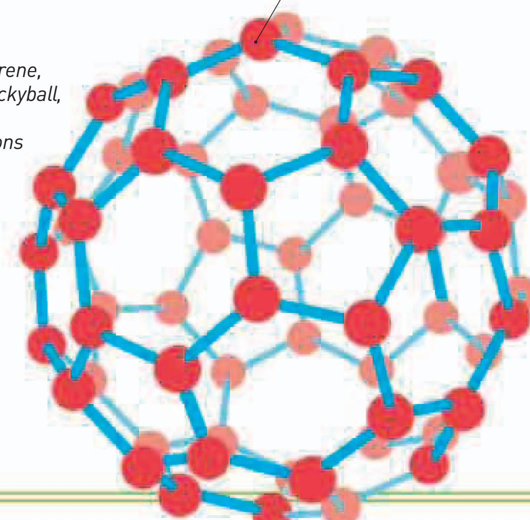
In May, scientists from the British Antarctic Survey (BAS) announced they had discovered a **downward trend in the concentration of ozone** above the Antarctic. Most atmospheric



Alzheimer's protein
A beta-amyloid protein molecule shows how the twisted structure of these molecules helps make them clump together, forming plaques.

ozone is found in a layer between 12 and 18 miles (20 and 30 km) above ground, and is greatest above the poles. The ozone layer plays a vital role in **protecting life on Earth,** blocking out potentially harmful ultraviolet radiation. Within two years, the main cause of the depletion of atmospheric ozone was confirmed: synthetic compounds called chlorofluorocarbons (CFCs), which were widely used in refrigerators and as propellants in aerosol cans (see 1986–87).

The **element carbon is extremely versatile**, forming chains and rings in countless compounds. Even when pure, this versatility is apparent: two of its best known allotropes (forms) are diamond, in which carbon atoms are arranged in tetrahedrons (see 1797), and graphite, in which they form flat planes of hexagons. In September of this year, chemists at Sussex University, UK, and Rice University, Texas, found **evidence of a stable allotrope of carbon** consisting of molecules with 60 atoms. The scientists quickly worked out the molecules' structure: the carbon atoms are joined in hexagons and pentagons. The structure is similar to that of a geodesic dome designed by American architect Richard Buckminster Fuller, so the new allotrope was named **buckminsterfullerene**. The new allotrope had been predicted by other chemists, and it has since been found occurring naturally. It spawned interest in an **important new class of materials**, called fullerenes (see 1990–91).



Buckyball
A buckminsterfullerene, also known as a buckyball, has a structure of alternating pentagons and hexagons, like the sections of a soccer ball.

60 THE NUMBER OF **CARBON ATOMS IN** EACH MOLECULE OF **BUCKMINSTERFULLERENE**

May 16 British Antarctic Survey reports a **declining concentration of ozone** above the Antarctic

June Scientists analyze beta-amyloid protein, implicated in the **development of Alzheimer's disease**

October The first **case involving DNA profiling**—an immigration dispute

October 18 American biochemist Marvin Caruthers announces a way of **building sequences of DNA** to order

November 14 Scientists announce the discovery of a **new form of carbon**—the buckminsterfullerene

November 20 Launch of version 1.0 of Microsoft's **Windows operating system**